

Technology Architecture: UjenziXform TIS Integration with KRA eTIMS

Document reference: UJX-TIS-KRA-ARCH-2026-001

Version: 3.3 | **Date:** 3 June 2026

Organisation: UjenziXform Solution | **Platform:** UjenziXform

Registered address: Baragetuny Plaza Left Wing 3rd Floor Room 10, Ronald Ngala Street, Eldoret. P. O. Box 4146 - 30100 Eldoret

Website: <https://ujenzi-pro.vercel.app/>

TIS product: UjenziXform Trader Invoicing System (TIS) v1.0.0 | **Solution type:** OSCU (primary)

1. Document metadata

Field	Value
Prepared for	Kenya Revenue Authority (KRA) — third-party integrator sandbox application
Platform	UjenziXform (construction materials marketplace, Kenya)
TIS product name	UjenziXform Trader Invoicing System (TIS)
Organisation name	UjenziXform Solution
Registered address	Baragetuny Plaza Left Wing 3rd Floor Room 10, Ronald Ngala Street, Eldoret. P. O. Box 4146 - 30100 Eldoret
Website	https://ujenzi-pro.vercel.app/
TIS product version	1.0.0
Solution type	OSCU (Online Sales Control Unit) — primary; VSCU initialization paths supported in code
Sandbox status	Not yet assigned — UjenziXform is applying for KRA eTIMS sandbox access
Target sandbox URL	<code>https://etims-api-sbx.kra.go.ke</code>
Certification status	Applicant / in development — seeking KRA sandbox access and subsequent integrator certification
Document version	3.3
Document date	3 June 2026
Document reference	UJX-TIS-KRA-ARCH-2026-001

Scope: This document describes **how UjenziXform will integrate with KRA eTIMS** once KRA assigns sandbox access to us as a **potential third-party TIS integrator**. It is a forward-looking architecture submission — not a report of live sandbox operations. For a one-page overview see `EXECUTIVE_SUMMARY.md` in this certification pack.

2. Purpose of this document

This document accompanies UjenziXform's **application for KRA eTIMS sandbox access** as a **potential third-party Trader Invoicing System (TIS) integrator**.

It explains:

- **What UjenziXform is** and why we require eTIMS integration
- **How we will connect to KRA eTIMS** after KRA issues sandbox credentials
- The **TIS software we have developed** in preparation (secure proxy, purchase-order mapping, fiscal receipt UI, audit, vendor onboarding)

We do **not** claim that KRA sandbox access or integrator certification has already been granted.

3. What UjenziXform is

UjenziXform is a web platform that connects builders, suppliers, and administrators in Kenya's construction materials marketplace.

User type	Role
Builders	Buy construction materials; create and accept purchase orders (POs)
Suppliers	Sell materials; hold catalog, pricing, and tax identity
Administrators	Operate the marketplace and TIS integrator tools

Typical transaction: a builder selects materials from a supplier, a **purchase order** is created and accepted, payment may follow (e.g. Paystack), and delivery workflows may apply. **KRA fiscal invoicing will be tied to the purchase order**, not to a separate internal invoice module.

The platform is designed so that the **eTIMS fiscal response** becomes the tax receipt for supplier sales once KRA sandbox testing and certification are complete.

4. What the TIS does

The **UjenziXform Trader Invoicing System (TIS)** will:

1. Collect **supplier tax identity** (KRA PIN, legal name, branch, device serial where required)
2. Map **purchase order line items** to KRA item codes and tax metadata
3. Submit **sales invoices** and **credit notes** to **KRA eTIMS** via REST (`POST /invoices`)
4. Store the **KRA/SCU response** (verification URL, control unit reference, JSON snapshot)
5. Display a **fiscal receipt with QR code** to buyers and suppliers
6. Log submissions for **admin review and retry**

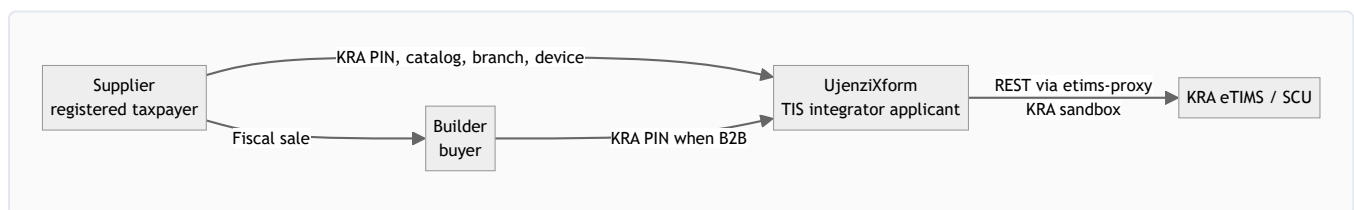
TIS is the **eTIMS compliance layer** embedded in UjenziXform order workflows — not a standalone accounting product.

5. Integrator application status

Item	Value
Applicant name	UjenziXform Solution
Registered address	Barnetuny Plaza Left Wing 3rd Floor Room 10, Ronald Ngala Street, Eldoret. P. O. Box 4146 - 30100 Eldoret
Platform	UjenziXform
Product name	UjenziXform Trader Invoicing System (TIS)
Product version	1.0.0
Solution type	OSCU
KRA eTIMS sandbox	Not yet assigned — applying for access
Target sandbox URL	<code>https://etims-api-sbx.kra.go.ke</code>
Target production URL	<code>https://etims-api.kra.go.ke</code> (only after KRA certification)
Integrator certification	Not held — applicant in development

UjenziXform is registering on the eTIMS portal and submitting this architecture as part of our request to be recognised as a **potential third-party TIS integrator** and to receive **KRA-assigned sandbox credentials**.

6. Parties and responsibilities

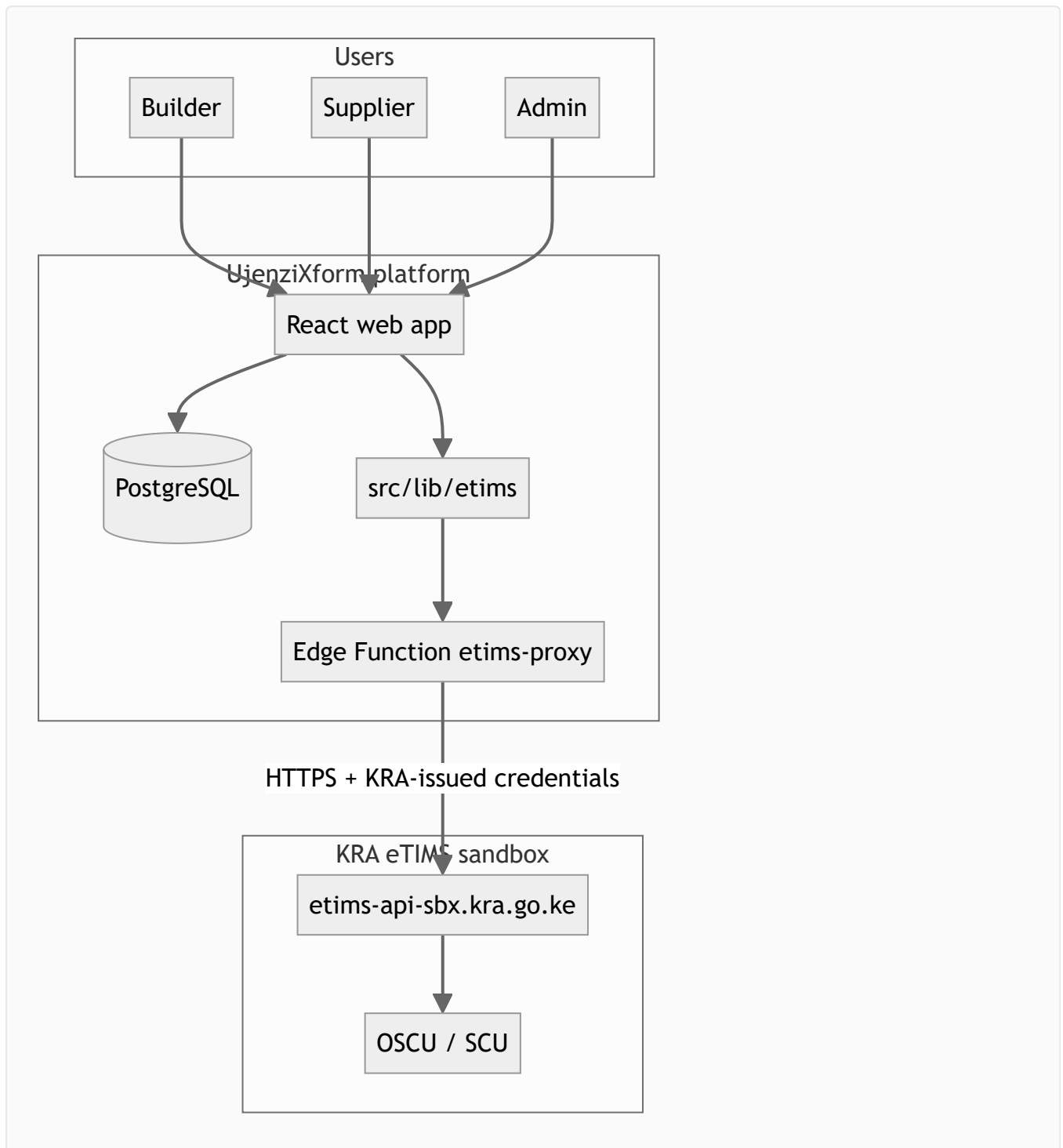


Party	Responsibility
KRA	Sandbox access, eTIMS policy, SCU, integrator approval
UjenziXform	TIS software, secure API gateway, vendor onboarding, audit
Supplier	Accurate KRA PIN, legal name, item codes; issues fiscal sale
Builder	Optional KRA PIN; receives fiscal receipt for verification

Each invoice will carry the **supplier's KRA PIN** as `traderTin` — the **vendor taxpayer** is the issuer, not UjenziXform's PIN alone.

7. Integration architecture

Once KRA assigns sandbox access, integration will work as follows:

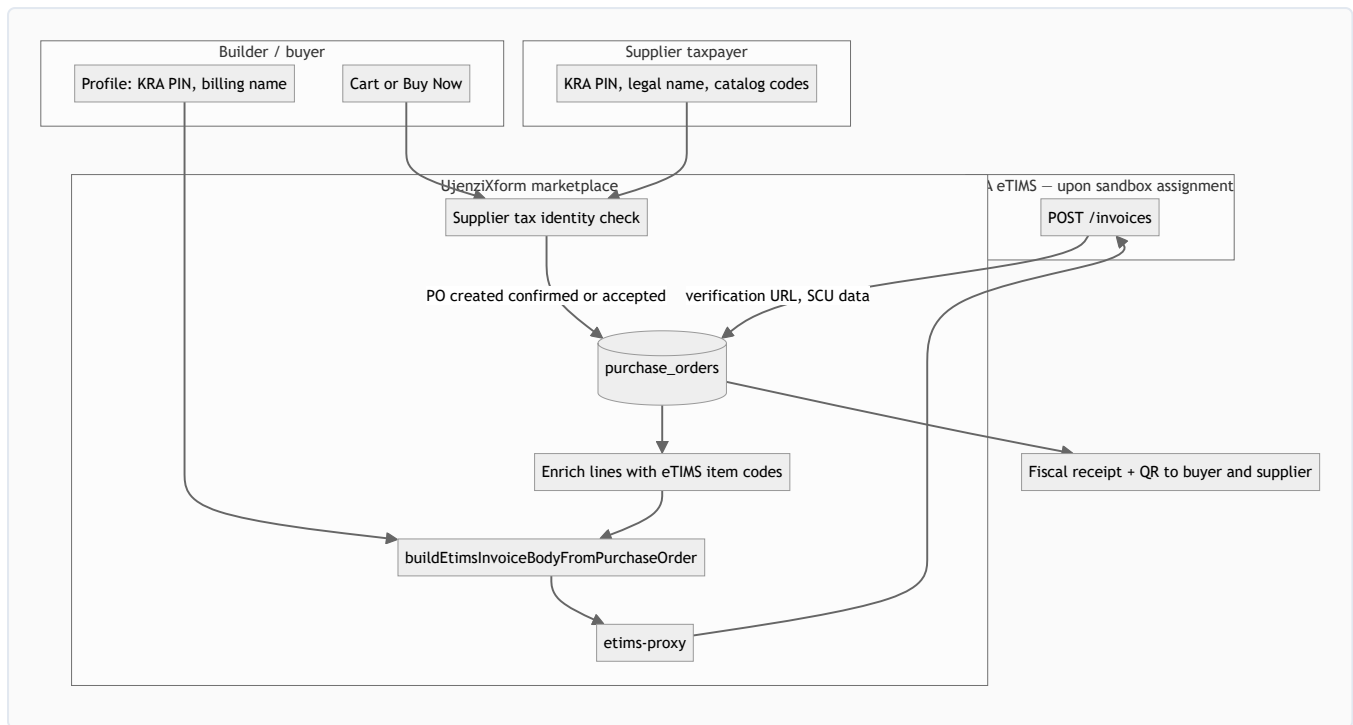


KRA-issued credentials will be stored as **server-side Edge secrets only**. The browser will never see them.

8. Marketplace purchase flow → POST /invoices

Yes — POST /invoices is part of the current UjenziXform purchase flow. TIS does not receive orders from outside the platform. A **purchase order (PO)** is the business document; TIS maps that PO to KRA **SalesReq** and submits it via `POST /invoices`.

This section describes the **current marketplace design**. Once KRA assigns sandbox access, the same flow will call **KRA eTIMS sandbox** instead of any other endpoint.



9. Purchase paths on UjenziXform

Path	What happens	Leads to POST /invoices?
Cart → Buy Now	Builder selects supplier, passes tax-identity check, PO created with status <code>confirmed</code>	Yes — automatic TIS submit after PO save (<code>CartSidebar</code>)
Materials grid → Buy Now	Direct single-item purchase, PO status <code>confirmed</code>	Yes — automatic TIS submit (<code>MaterialsGrid</code>)
Quote workflow	PO progresses through quote statuses until builder accepts	When accepted — PO must leave quote-pending statuses; supplier or admin can submit manually
Supplier Order Management	Supplier opens accepted PO	Yes — manual “Submit to KRA eTIMS” button
Admin TIS / eTIMS tools	Staff retry or test on a PO	Yes — manual submit / retry

Not wired today: automatic submit on quote accept (function exists in code but has no UI caller).

10. Gates before POST /invoices

Gate	Rule
Supplier tax identity	Valid <code>kra_pin</code> and legal business name (<code>assertSupplierTaxIdentityForCheckout</code>)
PO status	Blocked while <code>pending</code> , <code>draft</code> , or any <code>quote_*</code> / <code>quoted</code> status
Line item codes	Every PO line must resolve to an <code>etims_item_code</code> (from line JSON or catalog)
Buyer identity	<code>profiles.kra_pin</code> and billing name used as <code>customerPin</code> / <code>customerName</code> when present

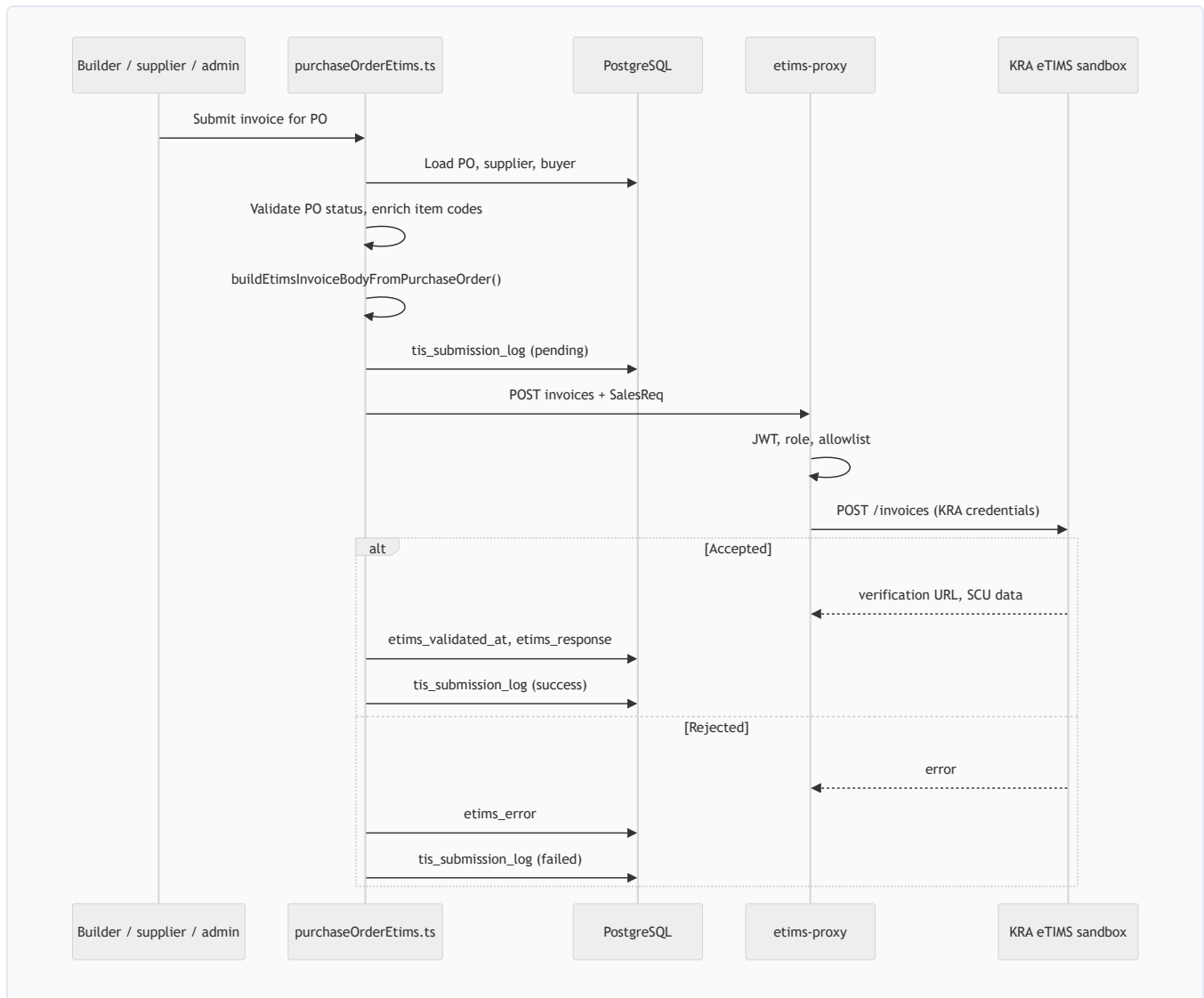
If a gate fails, TIS **does not call** POST /invoices; the error is stored on `purchase_orders.etims_error` .

The PO is the single source for the integrator payload: header fields (supplier KRA PIN as `traderTin` , buyer PIN/name, PO number as `traderInvoiceNo` , receipt type `s`) and `salesItems[]` from PO line JSON. Internal draft `invoices` rows on quote accept were **removed** (May 2026); the **eTIMS fiscal response** on the PO is the intended tax receipt.

11. Sales invoice flow — steps (KRA sandbox)

Step	Action
1	Supplier completes KRA PIN and legal business name
2	Catalog lines have KRA item codes
3	Builder's PO reaches accepted/confirmed status
4	TIS builds SalesReq from the PO
5	<code>etims-proxy</code> → POST <code>/invoices</code> on KRA sandbox
6	KRA returns SCU reference and verification URL
7	Response saved on PO; fiscal receipt + QR shown
8	Event logged in <code>tis_submission_log</code>

12. Sales invoice sequence diagram



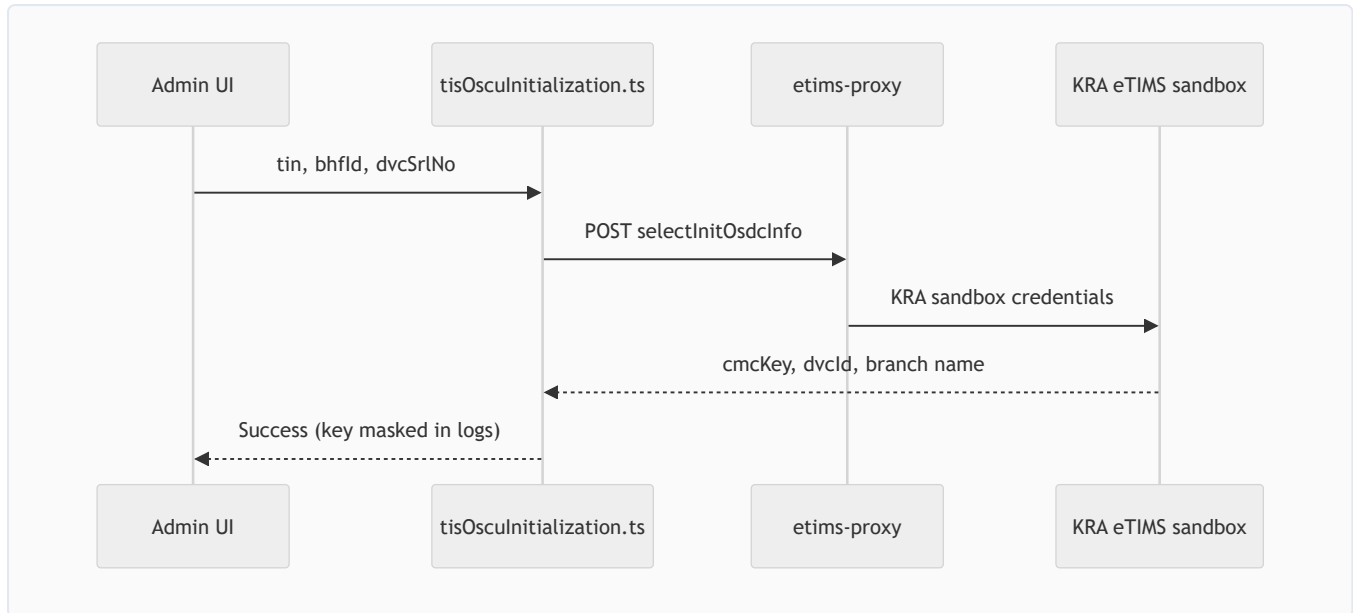
13. Submission triggers

Trigger	TIS design
Order confirmation / checkout	Automatic submit after PO confirmed
Supplier manual submit	Supplier dashboard action
Admin operations	Admin TIS tools for retry and testing

PO statuses that **will block** submission until accepted: `pending`, `draft`, `quote_created`, `quote_received_by_supplier`, `quote_responded`, `quote_revised`, `quote_viewed_by_builder`, `quoted`.

14. OSCU initialization flow

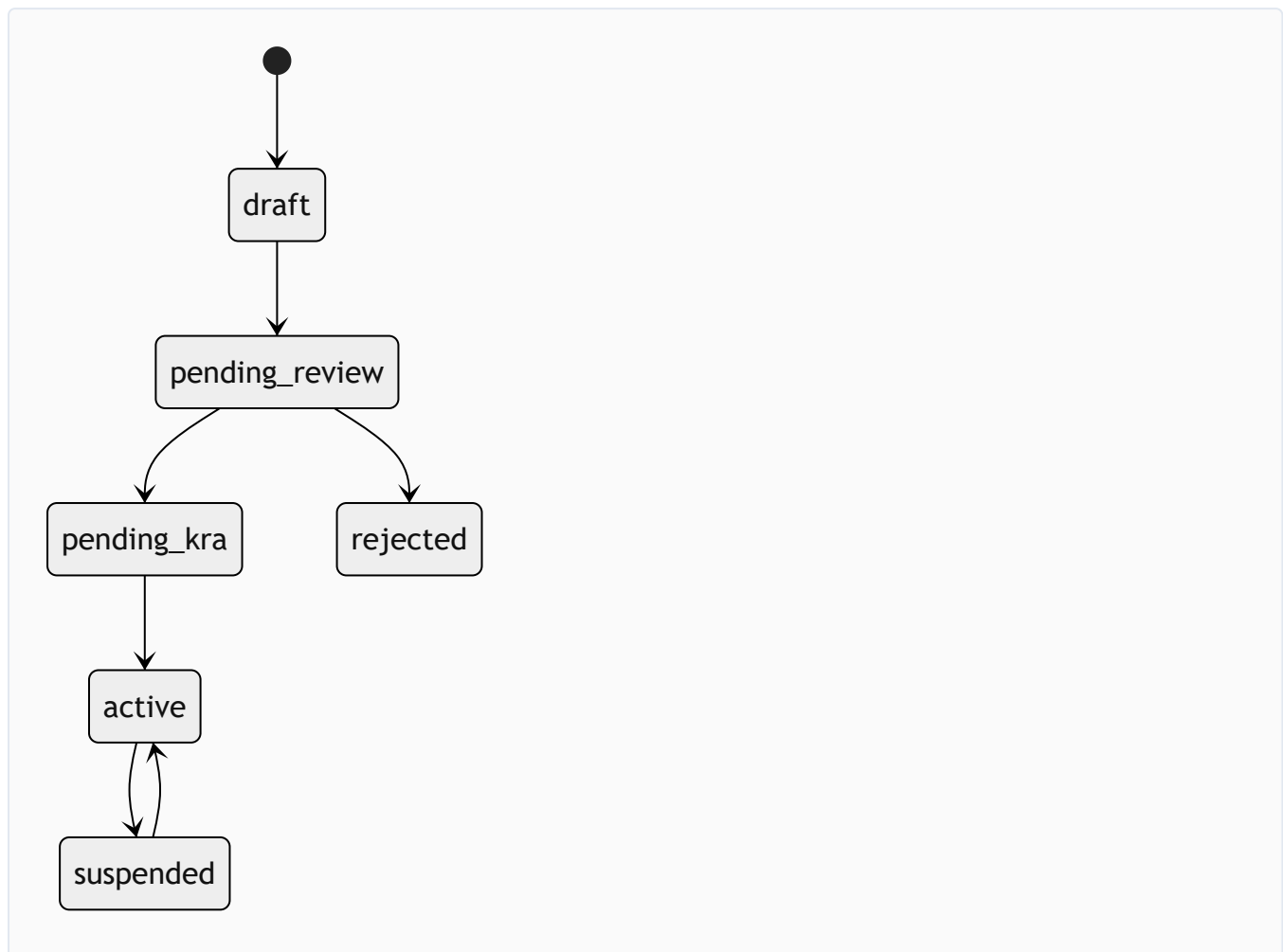
After KRA assigns sandbox access, vendor devices will be initialized via the admin **TIS Integrator Hub**:



Field	Meaning
tin	Vendor/supplier KRA PIN
bhfId	Branch code (e.g. 00)
dvcSr1No	KRA sandbox device serial

Communication keys will **never be stored in plain text** in the database.

15. Vendor (supplier) onboarding flow



Administrators will track each supplier in `tis_vendor_onboarding`. Status `pending_kra` covers KRA-related steps (device init, item registration) under UjenziXform's integrator application.

Before first sandbox invoice: valid supplier KRA PIN, legal name, item codes on all PO lines, buyer PIN when B2B.

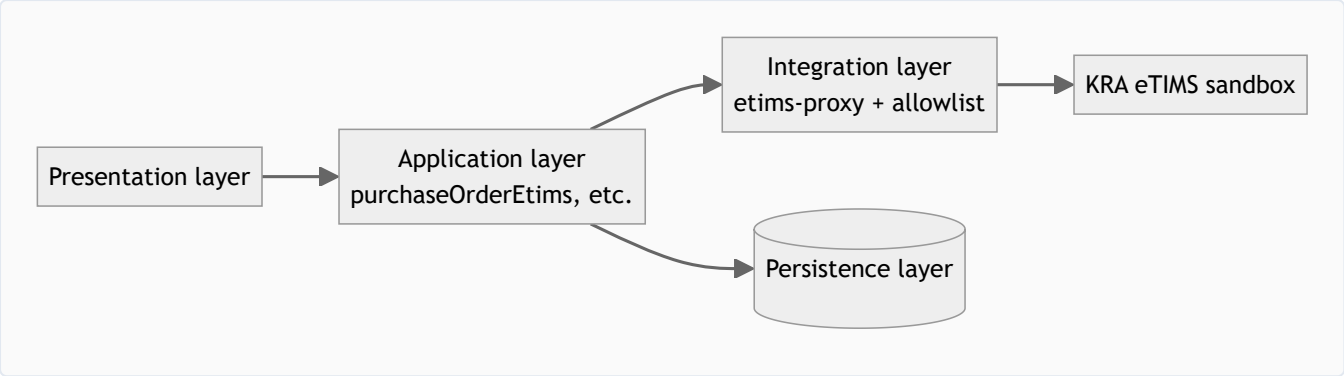
16. Credit notes

Credit notes will use the same `POST /invoices` endpoint on KRA sandbox with `receiptTypeCode: R` and `traderOrgInvoiceNo` referencing the original sale. Requires a prior successful sale on the PO.

17. Master data and catalog

Before sandbox sales, each PO line will require a KRA **item code** (from PO line, `supplier_product_prices`, `materials`, or admin catalog). Item registration (`POST /items`), customer registration (`POST /customers`), and reference-data sync will be performed against **KRA sandbox** through the allowlisted proxy and admin Integrator API console.

18. Technical architecture layers



Layer	Technology
Frontend	React 18, TypeScript, Vite
Backend	Supabase PostgreSQL, Auth, Row Level Security
Gateway	Supabase Edge Function <code>etims-proxy</code>
Client library	<code>src/lib/etims/*</code>

19. API integration (etims-proxy)

All outbound KRA traffic will use one Edge Function calling `{ETIMS_BASE_URL}/{path}` with HTTP Basic authentication.

Planned configuration after sandbox assignment:

Edge secret	Value
<code>ETIMS_BASE_URL</code>	<code>https://etims-api-sbx.kra.go.ke</code> (no trailing slash)
<code>ETIMS_BASIC_USER</code>	Issued by KRA
<code>ETIMS_BASIC_PASSWORD</code>	Issued by KRA

Allowlisted paths match KRA OSCU REST specification: initialization, reference data, items, customers, invoices, stock/purchase/import sync (`supabase/functions/_shared/etimsPathAllowlist.ts`).

Access controls: JWT required; roles `supplier` , `admin` , or `super_admin` ; SSRF path allowlist; 400 KB body limit.

20. Sales invoice payload mapping

Integrator field	UjenziXform source
traderInvoiceNo	PO number
traderTin / sellerPin	Supplier kra_pin
traderName	Supplier legal / company name
customerPin	Buyer profiles.kra_pin
salesItems[].itemCode	Catalog / PO etims_item_code
receiptTypeCode	S (sale) or R (credit note)
paymentType	Supplier default or 01
salesDate	yyyyMMddHHmmss at submission
currency	Default KES

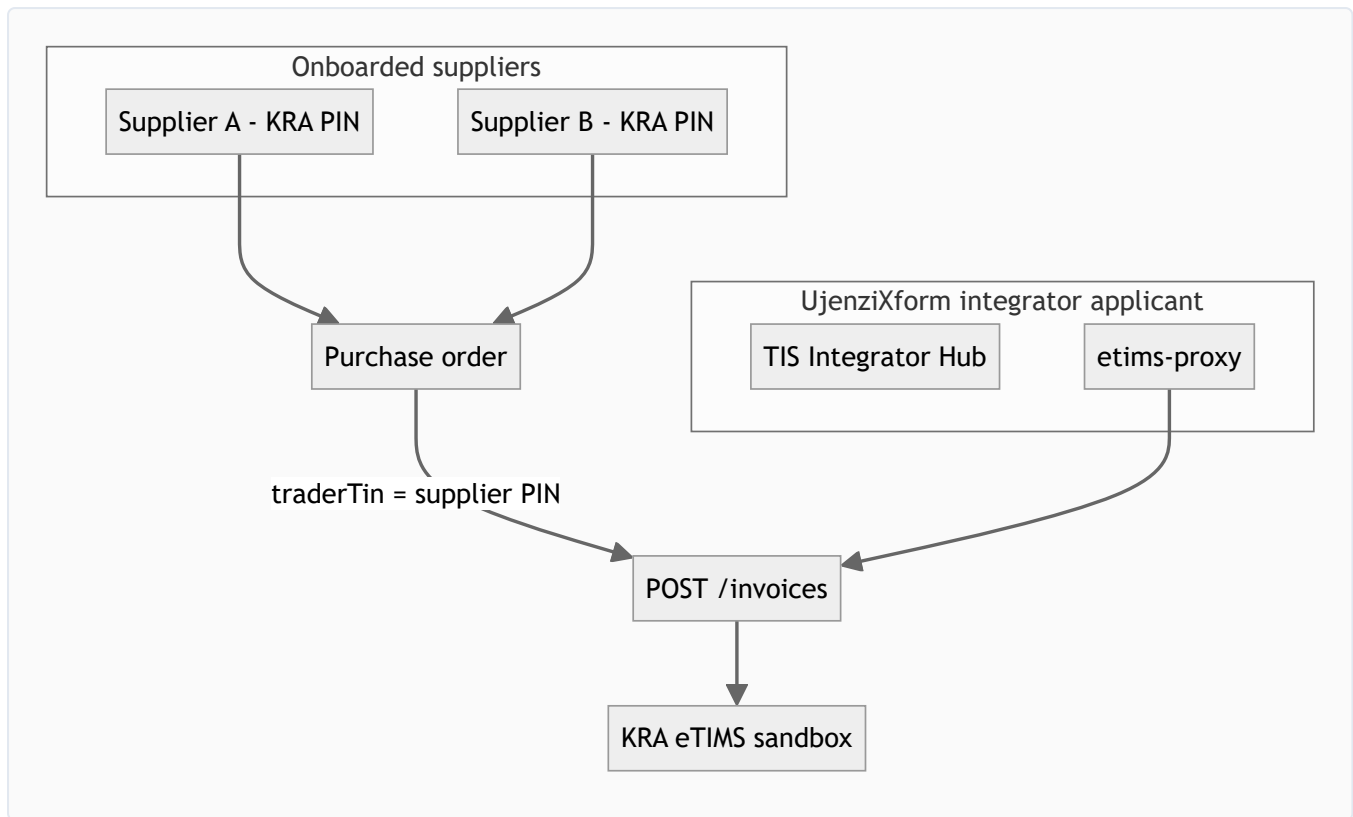
21. Data stored in the platform

Store	Purpose
purchase_orders (etims_*)	Fiscal submission outcome per order
suppliers	Vendor tax identity, branch, device serial
profiles	Buyer KRA PIN and billing identity
tis_integrator_platform	Integrator metadata and certification checklist
tis_vendor_onboarding	Per-supplier onboarding lifecycle
tis_submission_log	Submission audit trail

Not stored: KRA credentials, communication keys in plain text, secrets in browser code.

22. Multi-vendor (multi-taxpayer) model

UjenziXform will operate **one integrator registration** and serve **many supplier taxpayers** on the same marketplace. Each invoice will use that supplier's KRA PIN as `traderTin`.



23. Security controls

Control	Implementation
Credential isolation	KRA credentials in Supabase Edge secrets only
Authentication	Supabase JWT on every proxy call
Authorization	Roles: supplier, admin, super_admin
SSRF prevention	Strict path allowlist
RLS	TIS admin tables via <code>is_tis_integrator_staff()</code>
Key handling	OSCU communication keys redacted in audit logs
Request limit	400 KB max proxy body

24. Fiscal receipt shown to buyers

After a successful KRA submission, `EtimsFiscalReceiptView` will display:

- Issuer name and KRA PIN
- Trader invoice number and SCU-related fields
- Line items and totals
- **Verification URL** and **QR code** for KRA public validation

25. Audit, errors, and retry

Mechanism	Purpose
<code>tis_submission_log</code>	Per-submission audit (type, status, error, response)
<code>purchase_orders.etims_error</code>	Latest failure on the order
Admin Submission Ops	Staff retry and monitoring
Supplier dashboard	Manual submit and credit note

26. Post-sandbox-assignment plan

Once KRA issues sandbox credentials, UjenziXform will:

Step	Action
1	Configure Edge secrets with KRA-issued sandbox credentials
2	Set <code>ETIMS_BASE_URL</code> to <code>https://etims-api-sbx.kra.go.ke</code>
3	Set platform environment to <code>sandbox_testing</code>
4	Complete OSCU initialization with KRA sandbox device serials
5	Register test items and customers on sandbox
6	Execute sales invoice and credit note test cases
7	Complete internal certification checklist
8	Submit sandbox test results to KRA for integrator approval
9	Production cutover only after formal KRA certification

27. Implementation readiness matrix

Capability	Readiness	Notes
Secure <code>etims-proxy</code> gateway	Ready	Awaiting KRA sandbox URL and credentials
PO → sales invoice / credit note mapping	Ready	To be validated on KRA sandbox
Fiscal receipt + QR UI	Ready	
OSCU initialization flow	Ready	To run on KRA sandbox
Vendor onboarding workflow	Ready	
Submission audit logging	Ready	
Certification checklist tracking	Ready	Admin TIS Integrator Hub
KRA sandbox access	Pending	Subject of this application
KRA integrator certification	Pending	After successful sandbox testing
Production eTIMS	Future	After KRA approval

28. KRA certification checklist mapping

To be completed on **KRA-assigned sandbox** after access is granted:

ID	Requirement
platform_registered	eTIMS portal integrator registration
sandbox_access	KRA-issued sandbox credentials
oscu_vscu_init	Device initialization on KRA sandbox
master_data_sync	Reference data endpoints
item_registration	POST /items
customer_registration	POST /customers
sales_invoice	POST /invoices type S
credit_note	POST /invoices type R
fiscal_receipt_ui	Receipt + QR display
vendor_tis_identity	Per-supplier PIN, branch, device
vendor_onboarding_workflow	Onboarding lifecycle
submission_audit	Logging and retry
secure_credentials	Server-side Edge secrets
production_cutover	After KRA certification

29. Source code references

Component	Path
Edge proxy	supabase/functions/etims-proxy/index.ts
Path allowlist	supabase/functions/_shared/etimsPathAllowlist.ts
PO → invoice	src/lib/etims/purchaseOrderEtim.ts
OSCU init	src/lib/etims/tisOscuInitialization.ts
Audit log	src/lib/etims/logTisSubmission.ts
Admin hub	src/components/admin/tis-integrator/TisIntegratorHub.tsx
Platform DB	supabase/migrations/20260525120000_tis_integrator_platform.sql

30. Appendix — sample SalesReq payload

KRA reviewers often require a concrete example of the **POST /invoices** body. UjenziXform provides a separate appendix (anonymised, illustrative PINs and item codes):

File	Contents
appendix/APPENDIX_SAMPLE_SALESREQ.pdf	Print-ready appendix: HTTP call, field mapping, credit note variant, response shape, full sales invoice JSON (two PO lines)
appendix/APPENDIX_SAMPLE_SALESREQ.md	Source markdown for the appendix
appendix/APPENDIX_SAMPLE_SALESREQ.json	Machine-readable SalesReq JSON

Summary: A confirmed UjenziXform PO (e.g. `PO-2026-004821`) maps to SalesReq with supplier `traderTin`, buyer `customerPin`, and `salesItems[]` from catalog `etims_item_code` values. See **APPENDIX_SAMPLE_SALESREQ.pdf** in the certification pack for the complete payload and PO-field mapping table.

31. Document control

Version	Date	Changes
1.0	2026-06-03	Initial draft
2.0	2026-06-03	KRA presentation: business context, maturity matrix
3.0	2026-06-03	Sandbox application framing
3.1	2026-06-03	Application-only narrative
3.2	2026-06-03	Marketplace purchase flow → POST /invoices
3.3	2026-06-03	Executive summary + SalesReq appendix (separate files); comprehensive print-ready HTML architecture document

This document describes how UjenziXform will integrate with KRA eTIMS as a potential third-party TIS integrator after sandbox access is granted. For authoritative API contracts, refer to the official KRA eTIMS OSCU/VSCU Technical Specification.